Final Year Project Report Summary

Title: Computer Vision Based Parking Authorization and Authentication System

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1. Project Overview & Objective

The project was developed to replace traditional, single-factor parking gates with a smarter, more secure access control system. By combining physical hardware sensors with computer vision and a local database, the system handles entry and exit securely while keeping track of vehicles in real time. It mitigates common issues like tailgating and unauthorized token sharing without requiring complicated multi-camera setups.

2. Core Technological Stack

Tier	**Implemented Technologies & Tools**
Edge & Microcontroller	Arduino (C++), MFRC522 RFID transceivers (SPI bus), Servo motors (PWM), 4×4 matrix keypad, I2C display.
Software & Vision	Python, OpenCV, EasyOCR, multi-stage frame filtering, bilateral filtering, adaptive thresholding.
Database & Concurrency	Local SQLite relational database with atomic transactions and binary state tracking.
Generative AI & RAG	Google Gemini Flash, LlamaIndex, LlamaCloud, HuggingFace vector embeddings, Pinecone vector database.

3. Core Architectural Components & Methodology

Subsystem	Engineering Mechanism & Implementation
Dual-Factor Hardware Trigger	Wired twin MFRC522 RFID transceivers using an SPI bus to control entry and exit gates, turning physical card scans into automated software triggers.
Dynamic Optical Repositioning	Used a microcontroller-driven servo motor to shift a single camera's view between 0° (entry) and 180° (exit), saving costs over multi-camera setups.
Computer Vision & OCR Pipeline	Developed a Python backend with OpenCV and EasyOCR, adding frame-flushing, a 1.5x digital zoom, and bilateral filtering to handle motion blur and read license plates clearly.
Deterministic Anti-Passback Engine	Set up a local SQLite database that tracks an is_inside status for every user, making sure people can't pass cards back or tailgate.
Generative AI & RAG Assistant	Integrated Google Gemini Flash, LlamaIndex, and a Pinecone vector database so administrators can search project logs and docs using natural language.
Auditable Emergency Override	Built a 4 X 4 matrix keypad and hardware fail-safe loop to let staff trigger emergency gate openings safely, with all events saved to audit logs.

4. Key Engineering Results & Performance Metrics

Regarding engineering performance, the project achieved low-latency communication with an average serial transmission delay of under 2.5 milliseconds between the Arduino microcontroller and the Python backend. Algorithmic resilience also proved robust; the multi-stage image processing pipeline successfully cleaned up plate distortions and delivered high optical character recognition accuracy despite changing lighting conditions. Finally, rigorous load testing with multiple simulated users verified transactional integrity, demonstrating that the local database handles quick, back-to-back entries reliably without any data loss or tracking state anomalies.

5. Conclusion & Future Recommendations

In conclusion, the project demonstrated that combining low-cost microcontrollers with real-time computer vision and simple local databases offers a practical and secure alternative to traditional parking barriers, while the integration of an AI search assistant greatly streamlined log and documentation querying. For future development, migration from the split physical architecture to a unified edge computing appliance, such as an NVIDIA Jetson platform, is recommended to execute heavy neural vision and LLM tasks locally on a single board, alongside expanding support for multi-lane deployment.